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Monte Carlo Methods and Nuclear Modeling

Introduction to Monte Carlo Modeling

Problem 1

Under the guidance of Professor:

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The present document formatting has been aligned to meet standard practices as per ISO9001 in support of future standardization processes.

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1. Introduction.

The present document compiles class exercises as taught by lead “Monte Carlo Methods and Nuclear Modeling” teacher, Professor Christoph HARTNACK, with the intent of:

- Documenting resolution methodology,
- Annotate professor-student comments,
- Document lessons learned,
- Document relevant bibliography,
- Provide a quick entry point to tackle related daily life problems in future professional endeavors.

1.1. About the problem.

The problem prompts to find a minimum viable algorithm, preferably MatLab-Octave based, that would allow to implement Monte-Carlo analyses to estimate a participant-spectator analysis (e.g. <https://doi.org/10.1103/PhysRevLett.37.1202>) to assess fundamental nuclear phenomena.

1.2. Source code location and communications.

The Octave GNU source code developed to resolve this Problem 1 is publicly available at: https://raw.githubusercontent.com/J4v1987/MatLab-Octave_Nuclear_Modelling/refs/heads/main/Problem1.m

Should the formatting of plots and figures in this report exhibit suboptimal quality, do consider consulting flat 2D *.png, as well as interactive 3D *.ofig, from the same source code repository:

https://github.com/J4v1987/MatLab-Octave_Nuclear_Modelling/tree/945f312d651e37109ebec891d98a36f7933a736f/Problem1_Plots

Professionals in the nuclear engineering realm and interested audience are invited to participate in the development of this script. Do consider contacting the repository maintainer via dedicated communications found in:

<https://sites.google.com/view/b-eng-jarl/home>

2. Problem 1 enunciation

● Introduction to Nuclear Modelling, Master SNEAM

Student laboratories

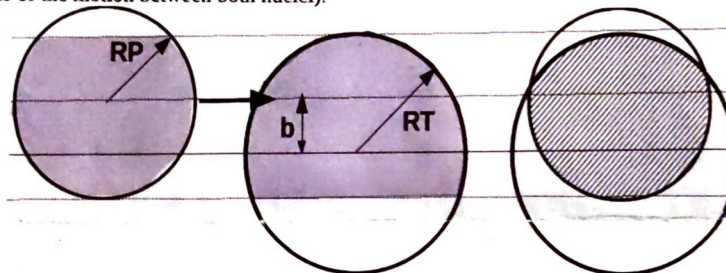
Christoph Hartnack

Problem 1 : Application of the Participant-Spectator-Model (evaluated)

In nuclear physics nuclei can be in first order regarded as spheres filled with nucleons. According to the liquid drop model the density inside the nucleus should be independent on the mass number A and thus the radius R directly related to A :

$$R = R_0 A^{1/3} \quad R_0 = 1.2 \text{ fm}$$

Imagine the collision of 2 nuclei with the mass numbers AP and AT . The impact parameter b describes the transverse distance of the 2 centers, i.e the distance perpendicular to the beam axis (the axis of the motion between both nuclei).



The participant-spectator model distinguishes the nucleons in the following way:

- all nucleons of the projectile/target which are also in the projection (perpendicular to the beam axis) of the other nucleus (target/projectile) are assumed to collide with nucleons of the other nucleus. They participate in the reaction and are thus called **participants**.
- all nucleons that do not fulfill this requirement may not take part in the reaction and are called **spectators**.
- The **participant fraction** is the ratio of the number of participants to the number of all nucleons. In our model we can take this as the ratio of the volume of the participant matter to the volume of both nuclei.

Calculate the participant fraction as a function of impact parameter for the following systems:

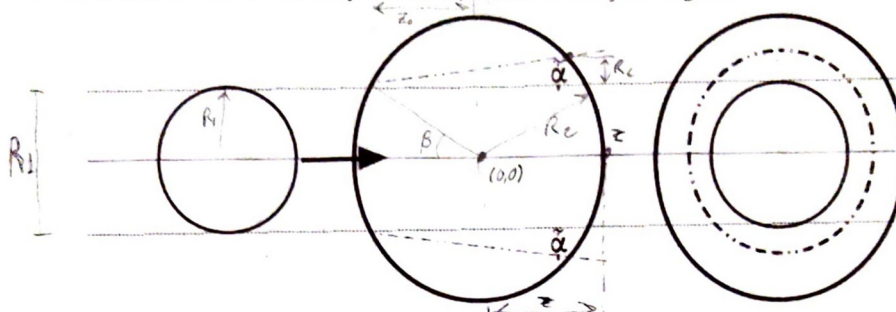
1. C+Au, $AP=12$, $AT=197$
2. Au+Au $AP=AT=197$

Your work should include the following steps:

1. Calculate the radii of the nuclei and determine the maximum impact parameter
2. Determine separately the conditions
 1. For a nucleon of the projectile to be a participant
 2. For a nucleon of the target to be a participant
3. Write a code that calculates the volumes of projectile, target, the projectile participants and the target participants. Calculate the participant fraction as function of b .
4. Plot the participant fraction as function of b for both systems C+Au and Au+Au
5. Discuss your results. Are there special cases with analytic solutions which may help to test whether your code works properly?

BONUS:
Extension of the participant-spectator model.

In the extension of this simple geometrical model one may imagine that the violent reactions of the participants create a sideways pressure which may extend into the spectator region. In this case some kind of cone may be created, characterised by the angle α .



For simplicity we will only regard central $b=0$ collisions of the system C+Au.

Calculate the participant fraction as a function of the angle α .

Which is the maximum "reasonable" value for α ?

Discuss your results. Are there special cases that help to test your code?

$$\frac{R_2}{z-z_0} = \tan \alpha$$

$$R_2 = R_1 + (z-z_0) \tan \alpha$$

$$\sin \beta = \frac{R_1}{R_2}, \quad \cos \beta = \frac{-z_0}{R_2}$$

$$\beta = \arcsin(R_1/R_2)$$

$$z_0 = -R_2 \cdot \cos(\beta)$$

3. Resolution

3.1. Octave usage (Case: C+Au)

```
>> clear all
>> close all
>> clc
>> Problem1
Projectile atomic mass [u] or [g·mol-1]: 12
Target atomic mass [u] or [g·mol-1]: 197
Projectile-target impact parameter scaling factor (100: just disengaged, 0: center to center):
80
Number Monte Carlo trials (e.g. 500): 10000
Do you want to see the analysis 3D plot? (may take longer to process)(yes or no) yes
```

PROCESSING...

3.2. Octave terminal outputs (Case: C+Au)

3.2.1.[00. ATOMIC MASSES]

Ap: Projectile particle atomic mass: 12 [u]
At: Target particle atomic mass: 197 [u]

3.2.2.[01. NUCLEI RADII & MAXIMUM IMPACT PARAMETER]

$\therefore R = R_0 \cdot A^{1/3}$ (Kenneth S Krane. ISBN: 0-471-80553-X, page 49.)
 $\wedge R_0 = 1.20$ [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 49.)
 $\wedge A = 12.00$ [u] (user input, script variable Ap)
 $\Rightarrow R(A_p) = R_p = 2.75$ [fm]

$\therefore R = R_0 \cdot A^{1/3}$ (Kenneth S. Krane. ISBN: 0-471-80553-X, page 49.)
 $\wedge R_0 = 1.20$ [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 49.)
 $\wedge A = 197.00$ [u] (user input, script variable Ap)
 $\Rightarrow R(A_t) = R_t = 6.98$ [fm]

$\therefore$ "...the impact parameter (the perpendicular distance from the center of the target nucleon to the line of flight of the incident nucleon). See: Kenneth S. Krane. ISBN: 0-471-80553-X, page 86.
 $\rightarrow b = R_p + R_t$ (maximum impact parameter, b, equals the projectile radius, R_p , plus the target radius, R_t .
 $\Rightarrow b = 9.73$ [fm] max.

Note: based in user prompt, b has been scaled to 80.00 %, where 0% denotes center-to-center impact, whereas 100% denotes maximum impact parameter b (projectile particle just grazed target particle).

The effective impact parameter in the analysis is: $b = 7.78$ [fm]

3.2.3.[02. PARTICIPANT-SPECTATOR FRAMEWORK]

While considering the "Nuclear Fireball Model" (see: <https://doi.org/10.1103/PhysRevLett.37.1202>), where the projectile particle collides through-all in straight trajectory line with the target particle, it then follows that "...The swept-out nucleons from the projectile and target are called the fireball...", where the fireball would constitute the participant nucleons, while the spectators would be the remainder nucleons.

Let there be a right-handed Cartesian coordinate system in standard Euclidean space as per ISO 80000-2:2009(E), E^n :

- The target particle of radius $R_t = 6.98$, is envisioned still, with center at (0,0,6.98)
- The projectile particle of radius $R_p = 2.75$, is envisioned as displacing at sub-relativistic speed along E^n x-axis. For Monte-Carlo analysis purposes it is envisioned to be in an instant of time with its center at (0,0,14.77).
- This array allowed to randomly scatter $N = 10000.00$ points within a prismatic volume tightly enveloping both the projectile and the target particles.

3.2.4.[03. VOLUMETRIC ANALYSIS]

ANALYTIC VOLUMES:

Monte Carlo test volume: 489.14 [fm³]

Projectile volume: 86.86 [fm³]

Target volume: 1425.93 [fm³]

MONTE CARLO POINTS:

Monte Carlo test points (encompassing the volumes of projectile and target), N: 10000.00 [points]

Projectile monte carlo points (assumed to be proportional to all projectile nucleons): 266.00 [points]

Target monte carlo points (assumed to be proportional to all target nucleons): 4205.00 [points]

PARTICIPANT POINTS:

Projectile participant monte carlo points: 82.00 [points]

Target participant monte carlo points: 119.00 [points]

PARTICIPANT FRACTION (Impact parameter, $b=7.78$ [fm] implicit):

Let:

Participant fraction: N_p

Target participant Monte Carlo points: $TPP=119.00$ [points]

Projectile participant Monte Carlo points: $PPP=82.00$ [points]

Projectile total Monte Carlo points: $PMP=266.00$ [points]

Target total Monte Carlo points: $TMP=4205.00$ [points]

It follows that:

∴ "...The participant fraction is the ratio of the number of participants to the number of all nucleons..." (C. Hartnack, "Student Laboratories, Master SNEAM, Application of the Participant-Spectator-Model, Problem 1")

→ $N_p = (TPP+PPP)/(PMP+TMP)$

⇒ **$N_p = 0.04$**

3.2.5. [04. Plots (C+Au)]

Projectile: Carbon (C) in blue-cyan.

Target: Gold (Au) in red-magenta.

3.2.5.1. (C) Projectile Points

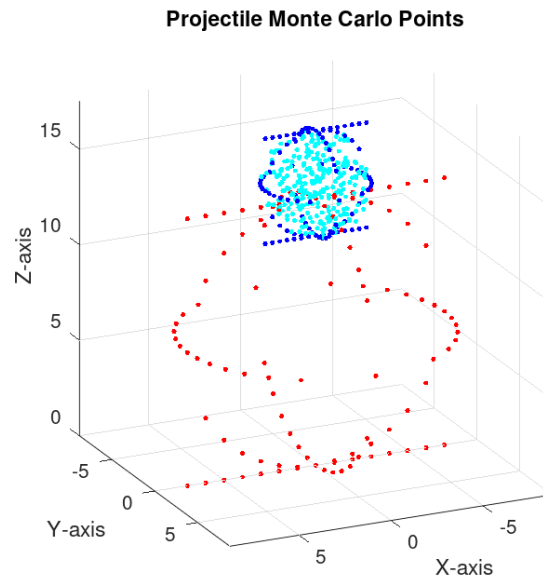


Figure 1: Carbon projectile Monte Carlo points

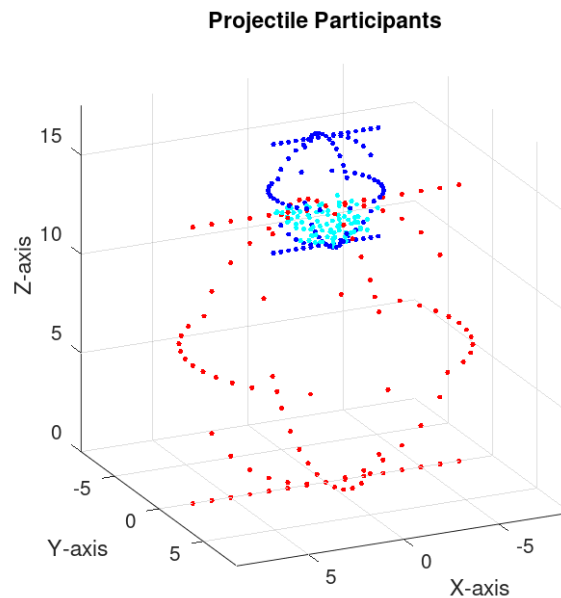


Figure 2: Carbon projectile participant points

3.2.5.2. (Au) Target Points

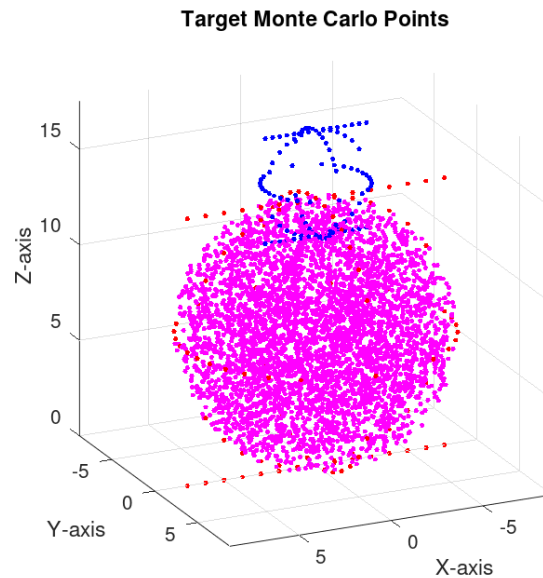


Figure 3: Carbon target Monte Carlo points

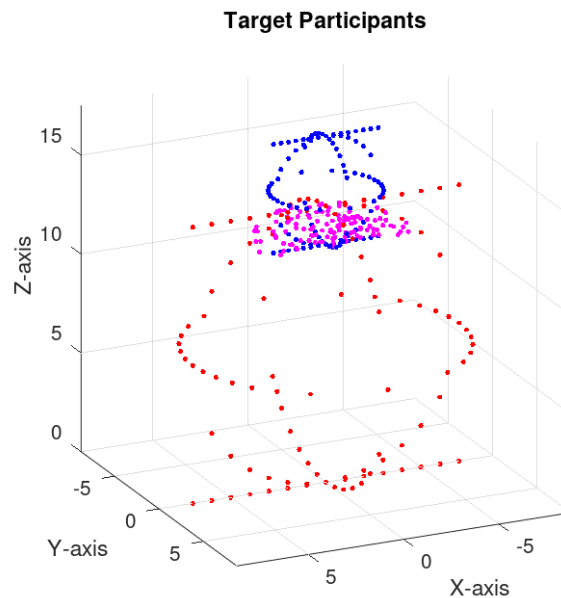


Figure 4: Carbon target participant points

3.2.6. [05. C+Au Results Discussion]

- Octave outputs matched in proportion the dimensions of the nucleons.
- Further source code trials may involve cases where:
 - Projectile is fully embedded into the target (e.g. $b=0$).
 - Target is smaller than the projectile.

3.3. Octave usage (Case: Au+Au)

```
>> clear all
>> close all
>> clc
>> Problem1
Projectile atomic mass [u] or [g·mol-1]: 197
Target atomic mass [u] or [g·mol-1]: 197
Projectile-target impact parameter scaling factor (100: just disengaged, 0: center to center):
80
Number Monte Carlo trials (e.g. 500): 10000
Do you want to see the analysis 3D plot? (may take longer to process)(yes or no) yes
```

PROCESSING...

3.4. Octave terminal outputs (Case: Au+Au)

3.4.1.[00. ATOMIC MASSES]

Ap: Projectile particle atomic mass: 197 [u]
At: Target particle atomic mass: 197 [u]

3.4.2.[01. NUCLEI RADII & MAXIMUM IMPACT PARAMETER]

∴ $R = R_0 \cdot A^{1/3}$ (Kenneth S Krane. ISBN: 0-471-80553-X, page 49.)
∧ $R_0 = 1.20$ [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 49.)
∧ $A = 197.00$ [u] (user input, script variable Ap)
⇒ $R(A_p) = R_p = 6.98$ [fm]

∴ $R = R_0 \cdot A^{1/3}$ (Kenneth S. Krane. ISBN: 0-471-80553-X, page 49.)
∧ $R_0 = 1.20$ [fm] (Kenneth S. Krane. ISBN: 0-471-80553-X, page 49.)
∧ $A = 197.00$ [u] (user input, script variable Ap)
⇒ $R(A_t) = R_t = 6.98$ [fm]

∴ "...the impact parameter (the perpendicular distance from the center of the target nucleon to the line of flight of the incident nucleon). See: Kenneth S. Krane. ISBN: 0-471-80553-X, page 86.

→ $b = R_p + R_t$ (maximum impact parameter, b , equals the projectile radius, R_p , plus the target radius, R_t).
⇒ $b = 13.96$ [fm] max.

Note: based in user prompt, b has been scaled to 80.00 %, where 0% denotes center-to-center impact, whereas 100% denotes maximum impact parameter b (projectile particle just grazed target particle).

The effective impact parameter in the analysis is: $b = 11.17$ [fm]

3.4.3.[02. PARTICIPANT-SPECTATOR FRAMEWORK]

While considering the "Nuclear Fireball Model" (see: <https://doi.org/10.1103/PhysRevLett.37.1202>), where the projectile particle collides through-all in straight trajectory line with the target particle, it then follows that "...The swept-out nucleons from the projectile and target are called the fireball...", where the fireball would constitute the participant nucleons, while the spectators would be the remainder nucleons.

Let there be a right-handed Cartesian coordinate system in standard Euclidean space as per ISO 80000-2:2009(E), E^n :

- The target particle of radius $R_t = 6.98$, is envisioned still, with center at $(0,0,6.98)$
- The projectile particle of radius $R_p = 6.98$, is envisioned as displacing at sub-relativistic speed along E^n x-axis. For Monte-Carlo analysis purposes it is envisioned to be in an instant of time with its center at $(0,0,18.15)$.
- This array allowed to randomly scatter $N=10000.00$ points within a prismatic volume tightly enveloping both the projectile and the target particles.

3.4.4.[03. VOLUMETRIC ANALYSIS]

ANALYTIC VOLUMES:

Monte Carlo test volume: 702.05 [fm³]

Projectile volume: 1425.93 [fm³]

Target volume: 1425.93 [fm³]

MONTE CARLO POINTS:

Monte Carlo test points (encompassing the volumes of projectile and target), N: 10000.00 [points]

Projectile monte carlo points (assumed to be proportional to all projectile nucleons): 2864.00 [points]

Target monte carlo points (assumed to be proportional to all target nucleons): 2943.00 [points]

PARTICIPANT POINTS:

Projectile participant monte carlo points: 219.00 [points]

Target participant monte carlo points: 234.00 [points]

PARTICIPANT FRACTION (Impact parameter, $b=11.17$ [fm] implicit):

Let:

Participant fraction: N_p

Target participant Monte Carlo points: $TPP=234.00$ [points]

Projectile participant Monte Carlo points: $PPP=219.00$ [points]

Projectile total Monte Carlo points: $PMP=2864.00$ [points]

Target total Monte Carlo points: $TMP=2943.00$ [points]

It follows that:

∴ "...The participant fraction is the ratio of the number of participants to the number of all nucleons..." (C. Hartnack, "Student Laboratories, Master SNEAM, Application of the Participant-Spectator-Model, Problem 1")

→ $N_p = (TPP+PPP)/(PMP+TMP)$

⇒ $N_p = 0.08$

3.4.5. [04. Plots (Au+Au)]

Projectile: Gold (Au) in blue-cyan.

Target: Gold (Au) in red-magenta.

3.4.5.1. (Au) Projectile Points

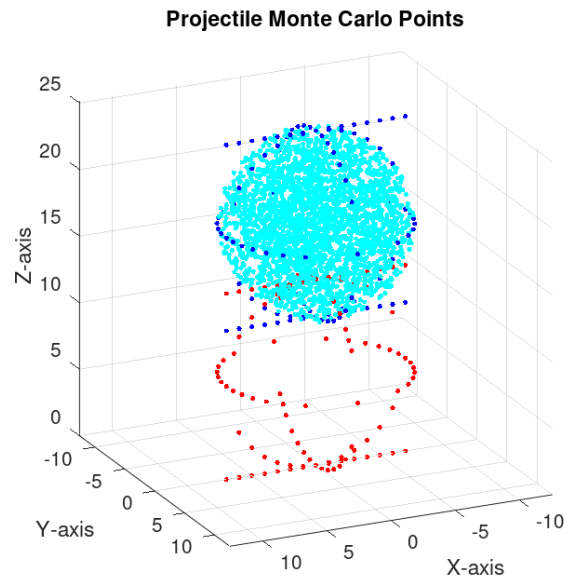


Figure 5: Gold projectile Monte Carlo points

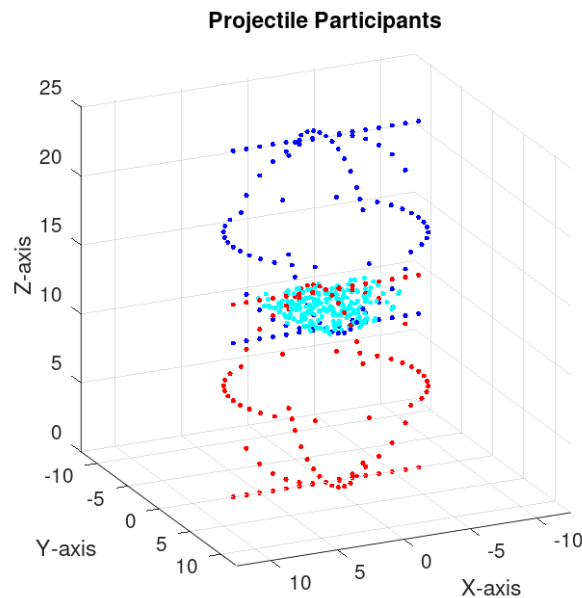


Figure 6: Gold projectile participant points

3.4.5.2. (Au) Target Points

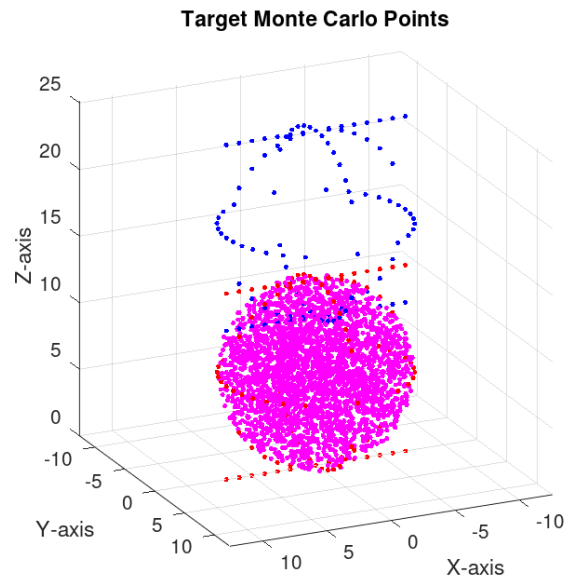


Figure 7: Gold target Monte Carlo points

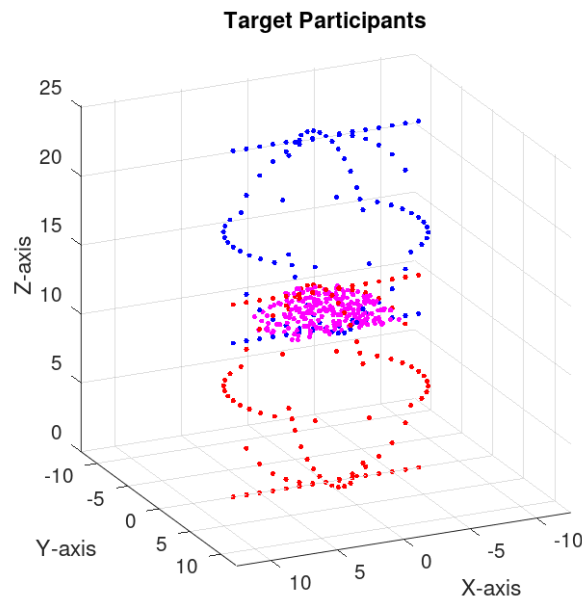


Figure 8: Gold target participant points

3.4.6. [05. Au+Au Results Discussion]

- Octave outputs matched in proportion the dimensions of the nucleons, but:
 - Although both projectile and target were of the same material (Au), Monte Carlo point counts varied slightly
 - The aforementioned is the result of the statistical nature of Monte Carlo methods, relying on random number generators.
 - It would be best practice to re-iterate the same set-up more than 30 times to obtain reliable standard deviations and thus assert statistically significant numbers.
- Further source code trials may involve cases where:
 - Projectile is fully embedded into the target (e.g. $b=0$).
 - Target is smaller than the projectile.

4. BONUS

One way to extend the participant-spectator model would be to envision the sideways pressure deforming the actors in through-all collisions. This would require to take into account material properties such as:

- Nucleon binding energies.
- Strong-forces.